DIGITALAIKART · INDIA EDITION

The Indian Diabetes & Prediabetes Playbook

2026-27

Your plate, your numbers and your next 90 days - built on Indian evidence, not imported advice.

101 million
adults in India live with diabetes

136 million
have prediabetes - and most don't know

7.7%
meet all three of their ABC targets

Facts verified November 2026 · ICMR-INDIAB · ICMR-NIN · RSSDI · IRDAI · MoHFW

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How to use this playbook
Read Part One once, properly, with your last lab report in front of you. Then go straight to Part Two and change one meal this week. Parts Three to Eight are reference: dip into the section you need rather than reading front to back. The worksheets in Part Eight are designed to be printed.

PART ONE

Where You Actually Stand

Before you change a single meal, you need to know what your reports mean, which numbers matter, and which ones Indian labs get wrong. Six sections, no jargon.

1. The two numbers that describe you

India has roughly 101 million adults living with diabetes and another 136 million with prediabetes. Those are the 2021 projections from the ICMR-INDIAB study, the largest nationally representative diabetes survey ever run in this country, published in The Lancet Diabetes & Endocrinology. The survey measured 113,043 people across 31 states and union territories and found a weighted diabetes prevalence of 11.4% and prediabetes at 15.3% among adults aged 20 and above.

Two details inside that study matter more than the headline. First, urban prevalence was 16.4% against 8.9% in rural India, but the gap is closing fast. Second, in Tamil Nadu, prevalence among adults rose from 11.1% in 2008-2010 to 22.7% in 2022-2023. That is a 104% increase in about a decade, with prediabetes rising 103% to 24.8%. The epidemic is not slowing down; it is accelerating in exactly the states that believed they had it under control.

And it is no longer a disease of the middle-aged. In the Longitudinal Ageing Study of India, among adults aged 45 and above, 30.4% had prediabetes and 49.8% had some form of dysglycaemia. Half of everyone over 45.

> **Why this matters to you personally**
> If you have been told you are 'borderline', you are in the largest single group in this country: 136 million people whose bodies are already struggling with glucose and who have not yet been labelled. What you do in the next three years decides which side of that line you spend the next thirty years on. That is the entire reason this playbook exists.

2. Reading your own report

Most people are handed a lab printout with six numbers and no explanation. Here is what each one actually tells you, in the order that matters.

Test	What it measures	What to look at
Fasting plasma glucose (FPG)	Glucose after 8-12 hours without calories. The cheapest, most widely available test.	The single most useful screening number in Indian settings.
2-hour plasma glucose (OGTT)	Glucose two hours after a 75 g glucose drink. Shows how well your body handles a sugar load.	Catches people whose fasting number looks fine but who spike after eating.
HbA1c	Average glucose over the previous 8-12 weeks, expressed as a percentage.	Your long-term grade. But read the HbA1c trap below before you trust it alone.
Fasting insulin / C-peptide	How much insulin you are producing. C-peptide shows your own pancreas output.	Tells you whether the problem is resistance or a failing pancreas. Changes the treatment conversation.

Test	What it measures	What to look at
Urine albumin-to-creatinine ratio (ACR)	Protein leaking into urine, a first sign of kidney strain.	The test most Indian clinics forget to order. Ask for it by name every year.
Lipid profile	LDL, HDL, triglycerides.	LDL below 100 mg/dL is a target, not a formality. It drives heart risk as much as sugar does.

Two practical warnings about Indian labs. Capillary (finger-prick) glucose is not a diagnostic test; the RSSDI guidelines are explicit that a diagnosis needs venous plasma. And plasma glucose in a whole-blood sample falls by 5-8% for every hour it sits at room temperature, so a sample drawn at a small lab and couriered across town can read falsely low. If a result surprises you in either direction, repeat it before you build a plan around it.

3. The diagnosis table, and the HbA1c trap

Indian guidelines do not perfectly agree with each other, which is why two doctors can look at the same fasting sugar of 104 mg/dL and say different things. Here is the honest picture.

Category	Fasting glucose	2-hour OGTT	HbA1c
Normal	Below 100 mg/dL	Below 140 mg/dL	Below 5.7%
Prediabetes (RSSDI recommended care)	110-125 mg/dL	140-199 mg/dL	5.7-6.4%
Prediabetes (ICMR standard treatment workflow)	100-125 mg/dL	140-199 mg/dL	5.7-6.4%
Diabetes	126 mg/dL or above	200 mg/dL or above	6.5% or above
Diabetes with symptoms	Random glucose of 200 mg/dL or above with classic symptoms is diagnostic on its own		

Notice the gap: the RSSDI's recommended-care threshold for impaired fasting glucose begins at 110 mg/dL, while the ICMR Standard Treatment Workflow for Type 2 Diabetes uses 100 mg/dL, the same as the American Diabetes Association. That single number is why a report can label you prediabetic at 104 and another doctor can call you normal. Neither is wrong; they are reading different documents. What matters is that a fasting glucose of 104 is not a number to ignore, whichever label you are given.

Now the trap. The RSSDI expert panel is unusually blunt about HbA1c in India: it should not be used as the sole criterion for diagnosing diabetes here. Iron deficiency anaemia, which is extremely common among Indian women, pushes HbA1c up without any real change in glucose. Haemoglobinopathies and thalassemia, more common in parts of the North East and in specific communities, distort it too. Studies cited in the guidelines found HbA1c readings significantly higher in people with iron deficiency anaemia (5.51% against 4.85%), and those readings fell after iron treatment alone.

The practical rule: ask for fasting glucose and HbA1c together, and add a 2-hour OGTT if either is borderline. If your HbA1c looks high but your fasting and post-meal glucose look reasonable, get your haemoglobin and ferritin checked before accepting a diagnosis. If your HbA1c looks reassuringly low but you have symptoms, do not stop there.

4. Prediabetes is a three-year window, not a life sentence

The most useful trial ever run on Indian bodies is the Indian Diabetes Prevention Programme, IDPP-1, published in Diabetologia. Researchers took 531 people with impaired glucose tolerance, all Asian Indians, average age 46, average BMI 25.8, and split them into four groups: control, lifestyle modification, metformin, and both together.

Over three years, 55% of the control group developed diabetes. In the lifestyle group, 39.3% did. That is a relative risk reduction of 28.5%, and a number needed to treat of 6.4, meaning you only have to change the trajectory of about six people with prediabetes to prevent one case of diabetes entirely. Metformin alone achieved 26.4%. Adding metformin to lifestyle changes did not beat lifestyle changes alone.

Group	Developed diabetes in 3 years	Risk reduction
Control (standard advice)	55.0%	-
Lifestyle modification	39.3%	28.5%
Metformin	40.5%	26.4%
Lifestyle + metformin	39.5%	28.2%

Read that again. In a population famous for being 'lean but insulin resistant', where the standard Western trial result was a 58% risk reduction, the Indian result was smaller but real, and it came from ordinary advice: fewer total calories, less refined carbohydrate, less fat, no sugar, more fibre, and a brisk 30-minute walk most days. Nothing exotic. Nothing expensive. No supplement.

That is the whole case for taking prediabetes seriously. You are not being asked to reverse a disease. You are being asked to not walk through a door that is currently open.

5. The ABC targets, and the 7.7% that should shame us

Diabetes care has three targets, not one, and India fails all three at scale. In the ICMR-INDIAB-13 analysis of 5,789 adults with known diabetes, 36.3% had good glycaemic control, 48.8% had controlled blood pressure, and 41.5% had controlled LDL cholesterol. Only 7.7% met all three together.

That is the single most important statistic in this book. It means the average Indian with diabetes is not failing because they are careless. They are failing because nobody told them there were three targets, and nobody is checking the other two.

Target	Standard goal	Notes
A - HbA1c	Below 7.0%	RSSDI and ICMR both use 7%. A tighter target may be set if it is achieved safely without hypoglycaemia; a looser one for frail elderly patients or those at high risk of lows.
B - Blood pressure	Below 130/80 (RSSDI); below 140/90 (ICMR workflow, 130/80 in kidney disease)	The two documents differ. Ask your doctor which applies to you, and know your own number.
C - LDL cholesterol	Below 100 mg/dL; below 70 mg/dL if you have heart disease	Statin therapy is usually needed to get there. Diet alone rarely does.
Pre-meal glucose	80-130 mg/dL	From the ICMR Standard Treatment Workflow.
Post-meal glucose	Below 180 mg/dL	Two hours after the start of a meal.

The other half of that study is about habits. Only 16.7% of people with diabetes used any self-monitoring of blood glucose at home, and even among those on insulin, only 36.9% did. Fewer than 25% did moderate-to-vigorous physical activity. Fewer than 20% ate even three servings of fruit and vegetables a day. So this is not a story about medicine failing. It is a story about nobody measuring, nobody moving, and nobody checking the other two letters of the alphabet.

6. Remission, reversal, and the words that sell books

You will see 'reverse diabetes in 21 days' advertised for Rs 4,999 on Instagram. Here is what the evidence actually supports.

Remission is a real, defined clinical state: an HbA1c below 6.5% sustained for at least three months without any glucose-lowering medication. It is most achievable in people with shorter duration of diabetes, lower starting HbA1c, and meaningful weight loss, and it is much more common in type 2 than most people realise. It is also not a cure. Beta-cell function has been damaged, and regaining weight reliably brings the glucose back.

What is not supported: that a specific food, powder, bitter gourd juice, karela-methi capsule, or 'alkaline water' reverses diabetes. That cinnamon replaces metformin. That you can stop prescribed medication because your readings look good for two weeks.

> **The one-sentence version**
> You can lower your HbA1c substantially, protect your eyes, kidneys and heart, and in some cases reach remission. You cannot do it by buying something. You can do it by changing what is on your plate, what you do after eating, and what you measure. Those three levers are what the rest of this book is about.

7. A worked example: one report, line by line

The values below are invented, for a 46-year-old man with a desk job who was told at a company health check that his sugar was 'slightly high'. Nothing here is a real patient. The point is to show you how to read the sheet in your hand rather than glancing at the one number your doctor circled.

Test	Result	Reference range printed	What it actually means
Fasting plasma glucose	118 mg/dL	70-100	Above normal, in the prediabetes band under the ICMR workflow, just short of the RSSDI's 110 cut-off but clearly abnormal. Needs a confirmatory test.
HbA1c	6.2%	Below 5.7	Consistent with prediabetes. Two tests agree, which is what you want before accepting a label.
2-hour OGTT	Not done	-	The test that would have shown whether he spikes after a sugar load. Worth asking for.
Total cholesterol	204 mg/dL	Below 200	Mildly high, and much less important than the LDL line below it.
LDL cholesterol	134 mg/dL	Below 100	The number that matters for heart risk. At 134 with prediabetes, this is a conversation about diet and possibly a statin.

Test	Result	Reference range printed	What it actually means
Triglycerides	196 mg/dL	Below 150	High. Often tracks with refined carbohydrate and alcohol intake, and with insulin resistance.
HDL cholesterol	38 mg/dL	Above 40	Low. Exercise raises it modestly; it is not something a pill fixes.
Blood pressure	138/88 mmHg	Below 120/80	Above target. In diabetes the goal is usually below 130/80, so this needs attention independent of sugar.
Serum creatinine	0.9 mg/dL	0.7-1.3	Normal, but normal creatinine does not exclude early kidney damage.
Urine ACR	Not done	-	The gap. Ask for it; it is the earliest sign of kidney involvement and it costs very little.
TSH	3.1 mIU/L	0.4-4.0	Normal. Worth checking once, because thyroid disease and diabetes travel together.
Vitamin B12	Not done	-	Relevant later if metformin is started.

Read as a whole, this report says something specific and useful: this man has prediabetes, with a lipid pattern and a blood pressure that already put his heart at higher risk, and nobody has yet looked at his kidneys. The single most valuable next step is not a supplement. It is an OGTT, a urine ACR, and a conversation about the 28.5% risk reduction that lifestyle change produced in the Indian prevention trial.

That is the difference between reading a report and receiving one. Six of the twelve lines above were never discussed in the consultation, and two of them were not even ordered.

> **The habit worth building**
> Every time you get a report, do three things before you file it. Circle anything outside the printed reference range. Write one sentence next to each circled number saying what it means. Take that page, not just the printout, to your appointment. You will get a different conversation.

PART TWO

The Indian Plate, Rebuilt

This is the heart of the playbook. Not a diet plan borrowed from a Western clinic, but the arithmetic of an Indian thali - what to cut, what to add, and why the swap you have been told to make may not work at all.

8. Why Indian diets break, in one study

In 2025, Nature Medicine published an analysis of dietary data from 18,090 adults in the ICMR-INDIAB survey. It is the most useful piece of nutrition research for Indians published in years, because it measured what people actually eat and then modelled what happens when you change it.

The findings, in plain language:

- Indian diets are 65-75% carbohydrate by calories, among the highest in the world.
- Of that, 62.3% of total energy comes from low-quality carbohydrate: white rice, milled whole grains, and added sugar.

- Protein intake is 9-11% of energy, well below the 10-15% that Indian guidelines recommend. Plant protein contributes 8.9%, dairy 2.1%, and animal protein just 1%.
- People in the highest carbohydrate intake group had a 30% higher odds of newly diagnosed type 2 diabetes than those in the lowest (odds ratio 1.30).
- 61% of the population was physically inactive.
- 83% of participants had at least one metabolic risk factor.

Now the part that should change how you shop. The researchers tested the swap almost every Indian doctor recommends: replace refined cereals with whole wheat or millet flour, keeping total carbohydrate the same. It did not reduce diabetes risk. The odds ratio was 0.94, with a confidence interval spanning 0.57 to 1.56, which is statistical language for 'no detectable benefit'.

Then they tested something different: replace carbohydrate calories with protein calories, keeping total calories the same. Every protein source worked. Substituting 5% of energy from carbohydrate with plant protein, dairy protein, egg protein or fish protein was associated with lower likelihood of type 2 diabetes and prediabetes, with odds ratios between 0.82 and 0.94.

> **The single most important idea in this book**
> Swapping your atta for millet atta while eating the same number of rotis changes almost nothing. Eating fewer rotis and adding a real protein source to the same plate changes a great deal. Most Indian 'diabetes diets' sell you the first and never mention the second. If you take one thing from these pages, take this.

This is not an argument against millets. Millets are nutritious, they carry more fibre and micronutrients than polished rice, and the ICMR-NIN Dietary Guidelines for Indians 2024 recommend them at 30-40% of your cereal intake. They are simply not a magic swap that cancels out quantity. Quality matters, and so does the total.

9. The 45/15 rule, translated into a thali

The ICMR-National Institute of Nutrition's Dietary Guidelines for Indians 2024 are the closest thing India has to an official national diet. Their numbers are more conservative than most people expect.

What	Official recommendation (ICMR-NIN 2024)
Cereals and millets	Not more than 45% of total calories. At least half of your cereals should be whole grain. Millets at 30-40% of the cereal portion by raw weight.
Pulses, beans, eggs, flesh foods	About 14-15% of calories combined. Actual Indian intake: 6-9%.
Carbohydrate overall	50-55% of calories
Protein overall	10-15% of calories
Fat overall	20-30% of calories, with visible fat under 30%
Added sugar	Under 5% of calories, roughly 25-30 g a day. Less is better.
Vegetables and fruit	Essentially half the day's plate. 400 g vegetables including green leafy vegetables, plus 100 g fruit in the 2,000 kcal plate.

The same document gives a worked 2,000 kcal day, which is genuinely useful because it is in grams rather than percentages. For a vegetarian: 250 g cereals and millets, 85 g pulses, 300 ml milk or curd, 400 g vegetables, 100 g fruit, 35 g nuts and seeds, 27 g of cooking oil. That totals roughly 55% carbohydrate, 15% protein and 30% fat.

Convert that to a kitchen. 250 g of raw cereal is about 8 to 9 rotis' worth of atta, or about 2.5 cups of raw rice. If you currently eat four rotis at lunch and four at dinner plus rice, you are already at or past the cereal ceiling for the whole day, before you count the poha at breakfast. That is the arithmetic almost nobody does.

Now the second number: 85 g of raw pulses is about 2.5 katoris of cooked dal. Most Indian households eat one. Doubling the dal and halving the rice is, in one sentence, the entire strategy.

10. Your plate, rebuilt

You do not need to weigh food for the rest of your life. You need a visual rule that survives a wedding buffet, a railway journey and your mother's kitchen. This is it.

Half of the plate	A quarter	A quarter
Non-starchy vegetables and greens - sabzi, salad, bhindi, lauki, beans, cabbage, palak, methi, cauliflower. Cooked in a teaspoon of oil, not floating in it.	Protein - dal, rajma, chana, paneer, curd, egg, fish, chicken. A visible portion, roughly the size of your palm.	Cereal - rice, roti, millet. Whatever fits in the last quarter, not the whole plate.

Three rules make this work in an Indian kitchen:

1. Vegetables go in first. Serve yourself the sabzi and salad before anything else. The plate that is already half full does not get filled with rice.
2. Protein must be visible. If you cannot point at the protein on your plate, it is not there. A thin dal is not a protein source; a katori of thick dal or a bowl of curd is.
3. Cereal is the last thing you add, not the base you build on. Invert the order you were taught.

There is a fourth, subtler rule that most people never hear: order matters within a meal. Eating the vegetables and protein before the rice or roti produces a lower glucose rise for the same food, because the fibre and protein reach the small intestine first and slow the emptying of the stomach. You are not changing what you eat, only the sequence. It costs nothing and it is one of the few free wins available.

11. Portions without a weighing scale

Kitchen measures you already own are more reliable than a half-remembered calorie app.

Food	One unit	Carbohydrate, roughly
Roti / chapati (medium, 30 g atta)	1 roti	18-20 g
Cooked rice	1 katori (150 g)	40-45 g
Cooked rice	1 large serving spoon	15 g
Idli	1 idli	12-15 g
Dosa (plain, medium)	1 dosa	25-30 g
Poha, cooked	1 katori	35-40 g
Upma, cooked	1 katori	35-40 g
Dal (thick, cooked)	1 katori	15-18 g
Rajma or chana (cooked)	1 katori	30-35 g
Curd	1 katori (150 g)	7-8 g

Food	One unit	Carbohydrate, roughly
Paneer	50 g	2-3 g
Chapati with ghee (1 tsp)	1 roti + ghee	20 g plus fat
Potato (boiled, medium)	1 potato	25-30 g
Banana (medium)	1 banana	25-30 g
Apple (medium)	1 apple	20-25 g
Tea with 2 tsp sugar	1 cup	8-10 g

These are approximations for planning, not laboratory values; brand, variety and cooking change them. The point is to give you a sense of scale. A single cup of rice is roughly the carbohydrate of two rotis. A plate of poha is roughly three rotis. That is why 'I only had poha for breakfast' and a reading of 210 mg/dL two hours later are not a contradiction.

One more number worth carrying in your head: 15 g of carbohydrate is one 'exchange'. A katori of rice is three exchanges. If your doctor or dietitian has given you a carbohydrate budget for a meal, this table lets you spend it without an app.

12. The carbohydrate ladder

Not all carbohydrates deserve the same place on your plate. Sort them into four rungs.

Rung	Foods	How often
Cut down hard	Sugar-sweetened drinks, fruit juice, packaged biscuits, cakes, sweets, white bread, maida-based snacks, jaggery and honey taken as 'healthy' sugar, flavoured yogurt, energy drinks	Rare, deliberate, small
Eat less of	White rice, maida roti, poha, upma, semolina, sabudana, potato in quantity, white-flour noodles, most restaurant breads	Portion-controlled, not the base of every meal
Eat normally, with portion sense	Whole wheat roti, brown or hand-pounded rice, oats, millets, corn, sweet potato, all pulses	At every meal, within the plate rule
Eat freely	Non-starchy vegetables, greens, salad, mushroom, beans, cucumber, tomato, brinjal, bottle gourd	Half the plate, every meal

A word on the middle rung, because it is where most confusion lives. Brown rice and white rice are close in glycaemic response when cooked soft and eaten in the same quantity; the fibre difference is real but modest. Hand-pounded or single-polished rice retains more bran and behaves better. Cold rice, reheated, has a little more resistant starch. None of these are transformations. They are refinements on top of portion control, and portion control is doing the heavy lifting.

13. Protein, the missing quarter

This is where Indian vegetarian households lose the most ground, and where the cheapest gains are available.

Source	Portion	Protein	Notes
Toor / moong / masoor dal	1 katori cooked	6-8 g	Thick dal, not the watery variety
Rajma, chana, lobia	1 katori cooked	8-10 g	Also high fibre; good for the plate's protein quarter
Paneer	50 g	9-10 g	Watch the fat if you fry it
Curd	1 katori	4-5 g	A protein and a ferment; useful at breakfast
Soya chunks (dry)	25 g	13-15 g	Cheapest high-quality protein in India
Egg	1 large	6-7 g	Whole egg; the yolk is not the enemy
Chicken	100 g cooked	25-28 g	Skinless
Fish	100 g cooked	20-22 g	Prefer baked, grilled or curry without deep frying
Milk	200 ml	6-7 g	Toned or double toned
Peanuts	30 g	7-8 g	Also fat; keep to a small handful

Two structural fixes that cost almost nothing. First, the 3:1 rule from the ICMR-NIN guidelines: combining cereals with pulses in a roughly 3:1 ratio improves the protein quality of the meal, because the amino acids each lacks are supplied by the other. Dal with rice is not a poor man's meal; it is a well-designed one. Second, put protein at breakfast. Most Indian breakfasts, poha, upma, paratha, bread, idli with chutney, are almost pure carbohydrate. Add two eggs, or a bowl of curd, or a besan chilla, and the same breakfast behaves completely differently for the next four hours.

How much protein do you need? The Indian guidelines put it at 10-15% of calories, which for a 2,000 kcal diet is about 50-75 g a day, and the official plate works out to roughly 72 g. Many people with diabetes do better at the higher end, closer to 1 g per kg of body weight, particularly if they are losing weight, because protein protects muscle while fat is lost. Discuss the number with your doctor if you have kidney disease; protein restriction applies there and it is a genuinely different conversation.

14. Fats, oils, and the ghee question

Fat is not the enemy, but Indian cooking has two specific problems: quantity, and the wrong kind.

Quantity first. The official plate allows 27 g of visible oil per day, which is about 5 teaspoons, for a 2,000 kcal diet. Measure what your household actually pours. A family of four deep-frying once a week, tempering three dishes a day and adding ghee to every roti can easily cross 60-70 g per person. Oil is 9 kcal per gram, so this is not a small difference.

Type second. Vanaspati, bakery shortenings, most commercial namkeen, biscuits, and repeated deep-frying oil carry trans fats, which raise LDL and lower HDL at the same time. Check labels for 'partially hydrogenated vegetable oil'; if it appears, put it back. Beyond that, rotating between mustard, groundnut, rice bran, olive and sesame oils is reasonable, and the ICMR-NIN guidance supports using different oils rather than one. Ghee is fine in small quantity, a teaspoon, as flavour rather than as a cooking medium; it is saturated fat and it counts.

Nuts deserve a specific mention because they are both excellent and easy to overdo. Thirty-five grams a day of mixed nuts and seeds is the official recommendation. That is roughly six to seven almonds plus two

walnuts, or a small handful of peanuts. A 'handful' that fills your palm is closer to 60-70 g and 400 calories.

15. Sugar, jaggery, honey, and the diabetic-food aisle

The ICMR-NIN 2024 guidelines recommend added sugar under 5% of calories, about 25-30 g a day, and note that if you eat sugar you must reduce cereals to compensate. That is the mechanism most people miss: sugar is not an add-on, it is a substitution. Two teaspoons of sugar in four cups of tea is 32-40 g of carbohydrate before you have eaten anything.

On the substitutes, plainly. Jaggery is sugar with trace minerals; it raises glucose the same way and is not a diabetes food. Honey is roughly 80% sugar and behaves like sugar. Brown sugar, coconut sugar and 'organic' sugar are sugar. None of them are free passes, and the health halo around them causes real harm because people use more.

The 'diabetic' and 'sugar-free' aisle is where the most money is wasted. A biscuit labelled diabetic-friendly is still a biscuit: refined flour, fat, and often more saturated fat than the regular version to compensate for texture. Sugar-free sweets contain sugar alcohols which can cause bloating and diarrhoea in quantity. Products sweetened with stevia or sucralose do reduce sugar, but they do not make the base food healthy.

> **Label rule in ten seconds**
> Ignore the front of the pack entirely. Turn it over and look at three things: total carbohydrate per 100 g, added sugar per 100 g, and the ingredient list order. If sugar, glucose, dextrose, maltodextrin, invert syrup or malt extract appears in the first three ingredients, it is a sweet, whatever the packaging claims.

16. Breakfast: the meal that sets your day

If you only fix one meal, fix this one. A high-carbohydrate breakfast produces a glucose spike that then drives hunger and snacking through the afternoon. Here are six common Indian breakfasts rebuilt, keeping the dish you already like.

You eat	The problem	Rebuild
Poha with peanuts	Flattened rice is refined carbohydrate; a katori is 35-40 g carb, and most people eat two	Half the poha, double the peanuts and peas, add a boiled egg or a bowl of curd on the side
Idli with coconut chutney	Rice plus rice; 3 idlis is 40 g carb with almost no protein	Two idlis, one bowl of sambar with extra dal, add a boiled egg or 50 g paneer
Aloo paratha with curd	Potato plus maida/atta plus ghee; a single paratha can be 45-55 g carb	One paratha, a much bigger bowl of curd, and a plain side salad; skip the butter
Bread, butter, jam	White bread, fat and sugar, with no protein at all	Two eggs on one slice of whole-grain bread, plus a fruit; drop the jam
Upma or vermicelli	Semolina is refined; low fibre, low protein	Add a cup of mixed vegetables and a handful of peanuts to the same upma, halve the portion, add curd
Only tea and biscuits	The worst option: sugar, refined flour, no protein, and you are hungry by 10 am	Tea with less sugar plus besan chilla, or eggs, or a bowl of curd with nuts

The pattern is always the same three moves: halve the cereal, add a protein, add a vegetable or a fruit. You do not need new recipes. You need different proportions in the recipes you already cook.

17. Ten lunches and dinners that work

Each of these is a real Indian meal, not a diet invention, built to the plate rule. Portions are for one adult; adjust for body size and activity, not for appetite.

Meal	What is on the plate	Why it works
Rajma chawal	1 katori rajma, half katori rice, big salad, 1 tsp ghee	The pulse carries protein and fibre; the rice is a quarter, not half
Dal-roti-sabzi	2 rotis, 1 katori thick dal, 1 katori seasonal sabzi, salad	Classic thali, correct proportions
Chicken curry with roti	100 g chicken, 2 rotis, 1 katori mixed vegetables	High protein, moderate carb
Fish curry with rice	100 g fish, half katori rice, 1 katori greens	Fish protein plus fibre; keep the gravy thin
Chole with one bhature	1 katori chole, one small bhatura (not two), onion salad	The chole is doing the work; the bhatura is a treat, not a staple
Khichdi with curd	1 katori khichdi with extra moong, 1 katori curd, 1 katori sabzi	Moong plus rice is a good amino acid pairing; curd adds protein
Paneer bhurji with roti	75 g paneer, 2 rotis, 1 katori salad	Protein-forward vegetarian dinner
Sambar rice with poriyal	Half katori rice, generous sambar with extra dal, 1 katori poriyal	Lentils and vegetables carry the meal
Egg curry with roti	2 eggs, 2 rotis, 1 katori beans or greens	Cheap, fast, high quality protein
Idli with sambar and a side	2 idlis, sambar with extra dal, 1 katori vegetable	Works when the sambar is thick and the vegetable is real

Note what is absent: nothing is banned. Rice appears five times. Roti appears four times. The meals differ from what most households eat only in the ratio of cereal to everything else.

18. The seven-day starter menu

If you want a plan rather than principles for the first week, use this. It is ordinary Indian food, cooked the way your household already cooks it, with the portions that fit an average adult eating roughly 2,000 calories a day. Adjust the cereal down if you are smaller or less active, and up if you are larger or physically working.

Day	Breakfast	Lunch	Dinner
Mon	2 besan chillas, 1 katori curd, tea without sugar	1 katori rajma, half katori rice, kachumber salad	2 rotis, 1 katori lauki sabzi, 1 katori dal
Tue	2 eggs or 50 g paneer bhurji, 1 slice whole-grain bread, 1 guava	2 rotis, 1 katori palak paneer, salad	1 katori moong khichdi, 1 katori curd, beans sabzi
Wed	1 katori poha with peanuts and peas, 1 katori curd	Half katori rice, sambar with extra dal, 1 katori poriyal	100 g chicken curry or 30 g soya chunks, 2 rotis, salad
Thu	2 idlis, thick sambar, 1 boiled egg	1 katori chole, 1 roti, onion salad	1 katori dal, 2 rotis, 1 katori bhindi
Fri	1 katori oats with milk, 10 almonds, tea without sugar	100 g fish curry, half katori rice, 1 katori greens	75 g paneer tikka, 1 roti, large salad
Sat	1 plain dosa, sambar, 1 katori curd	1 katori kadhi, half katori rice, 1 katori mixed vegetables	2 rotis, 1 katori rajma, salad

Day	Breakfast	Lunch	Dinner
Sun	2 eggs, 1 roti, 1 fruit	Your free meal - take half the usual cereal, full protein, walk 15 minutes after	1 katori dal, 1 roti, 1 katori sabzi

Snacks come from the list in the next section, one in the afternoon and one only if you are genuinely hungry. Water, buttermilk and unsweetened tea or coffee are free. Everything else in the week is a variation on this pattern, and after seven days you will not need the table.

Two notes on the Sunday free meal. It is not a cheat; it is a planned release valve that prevents the Friday-night collapse that undoes a fortnight. And it works only if it is one meal, eaten sitting down, followed by a walk. A free day is not a free weekend.

19. Your kitchen reset

Willpower is a poor strategy and a worse one at 5 pm. The reliable strategy is changing what is in the house. This takes one shopping trip and one Sunday afternoon.

Stop buying	Always keep	Buy less often
Biscuits, cream biscuits, 'digestive' biscuits	Atta, plus bajra, jowar or ragi flour	White bread
Namkeen packets, chips, bhujia	Three kinds of dal, rajma, chana, lobia	Maida
Fruit juice cartons and squash	Paneer, curd, milk, eggs, soya chunks	Vanaspati and bakery shortening
Sweetened yogurt and flavoured milk	Seasonal vegetables, greens, tomato, cucumber, onion, lemon	Refined oil in bulk
Jam, ketchup with sugar, sweetened sauces	Peanuts, almonds, walnuts, roasted chana, makhana	Ghee beyond a small tin
'Sugar-free' and 'diabetic' biscuits and sweets	Mustard or groundnut oil, plus one other oil to rotate	Potatoes in quantity
Sugar-sweetened soft drinks and energy drinks	Glucose tablets or glucose powder, for lows	Fruit in syrup, canned
Ready-to-eat gravies and frozen parathas	Tea, coffee, whole spices, ginger, garlic	Restaurant leftovers as next-day lunch

The Sunday ritual, about forty-five minutes: soak rajma or chana for the week and freeze in katori-sized portions; boil four eggs; portion nuts into five small bowls rather than leaving the packet open; chop a box of salad vegetables; cook one large pot of dal that will cover two dinners. The point of all of it is that the healthy option is also the easy option on a Tuesday night when you are tired.

20. Snacks that do not spike

Snacking is where good intentions dissolve, usually around 5 pm, usually standing up. Have a fixed list rather than a decision to make.

- A small bowl of curd with a few roasted chana.
- Two boiled eggs with salt and pepper.
- A handful of peanuts or roasted chana, measured into a bowl, not eaten from the packet.

- Besan chilla, one, made with a teaspoon of oil.
- A plate of cucumber, carrot and tomato with lemon and chaat masala.
- Paneer tikka, 50 g, grilled not fried.
- One fruit: guava, apple, orange, pear, or a small bowl of papaya. Not juice.
- Roasted makhana in a small bowl, dry-roasted not ghee-fried.
- A glass of buttermilk, unsalted or lightly salted.
- Sprouts chaat with onion and tomato.
- One slice of whole-grain bread with a boiled egg or 30 g paneer.
- A small bowl of dal, leftover from lunch, eaten like soup.

What to keep out of the house entirely, because willpower is not a strategy: biscuits, namkeen packets, chips, sweetened yogurt, fruit juice cartons, and anything in a box that can be eaten without a plate. If it is not in the kitchen at 5 pm, you will not eat it at 5 pm.

21. Eating out, weddings and street food

You cannot opt out of Indian social life, and you should not try. Fifteen rules that hold in a restaurant, a buffet and a thela.

1. Order one cereal, not two. Rice or roti, not both, unless you are sharing.
2. Ask for the gravy dishes with less oil where possible; many restaurants will do it.
3. Start with the salad or the soup, not the papad and the fried starter.
4. Tandoori, tikka, grilled, roasted and steamed are your friends. 'Crispy', 'golden fried' and 'butter' are not.
5. Eat the protein first, then the vegetable, then the rice or roti.
6. At a buffet, take one plate, fill half with salad and vegetables, and do not go back for the cereal section.
7. Skip the sweet drink. Nimbu pani without sugar, or water, or plain soda with lemon.
8. One sweet at a wedding, chosen deliberately, eaten after the meal rather than instead of it.
9. Do not arrive starving. A bowl of curd or a handful of nuts twenty minutes before you leave home changes what you order.
10. Street food: choose what is freshly cooked in front of you. Pani puri with the water, chaat without the fried puris, tikka without the maida roll.
11. Biryani: take the rice from the top rather than the bottom, where the ghee pools, and eat the meat and raita properly.
12. South Indian restaurants: two idlis or one dosa, extra sambar, and skip the potato masala filling where you can.
13. Alcohol, if you drink, is empty calories and it blunts the effect of your medication; keep it occasional and never on an empty stomach.
14. Finish eating within 20-25 minutes. Eating fast is independently associated with higher glucose and more intake.
15. Walk for ten minutes after you get home. It is the single best thing you can do after a heavy restaurant meal.

22. Festivals, vrats and fasting

A festival is not a reason to abandon a plan, and it is not a reason to be miserable either. The goal is to participate without losing a fortnight.

Diwali, and every festival built around sweets, is a two-week event, not a one-day one. Three practical rules: eat your sweet after a proper meal rather than instead of one; take one piece and put the box away rather than leaving it open on the table; and walk after the evening meal on every one of those days. A single kaju katli is about 10-12 g of carbohydrate. The problem is never the one piece; it is the tin, over eleven days.

On fasting, and this matters. If you take insulin or a sulfonylurea, fasting carries a real risk of hypoglycaemia and needs a conversation with your doctor before you begin, not during. Fasting rules that keep it safe: do not fast on consecutive days if you are on these medicines; keep taking metformin unless told otherwise; break the fast with protein and vegetables rather than a plate of puri and potato; and check your glucose if you feel shaky, sweaty, confused or unusually hungry.

Navratri and other vrat diets often replace cereals with sabudana, potato and makhana. Sabudana is nearly pure starch and behaves like sugar; a bowl of sabudana khichdi can be worse than the rice you replaced. Better vrat choices: paneer, curd, makhana, peanuts, rajma or chana where permitted, and fruit. If your tradition permits, add a bowl of curd to every vrat meal.

23. Tea, coffee, alcohol, tobacco and sleep

Four small levers that people underestimate.

Tea and coffee without sugar are neutral to mildly helpful; caffeine does not meaningfully raise glucose in most people. With two teaspoons of sugar and four cups a day, you are eating about 40 g of carbohydrate that satisfies nothing. Taper by half a teaspoon a week; taste adapts in about three weeks.

Alcohol lowers glucose acutely and can cause delayed hypoglycaemia overnight, particularly with insulin or sulfonylureas. If you drink, eat with it, never drink on an empty stomach, and check your glucose before bed.

Tobacco, smoked or chewed, independently worsens insulin resistance and multiplies cardiovascular and kidney risk in diabetes. It is the single highest-return change available to a smoker, larger than any dietary refinement in this book.

Sleep is the lever nobody mentions. Short or fragmented sleep raises insulin resistance the next day, and raises hunger. The RSSDI guidance for people with prediabetes suggests 6-8 hours. If you snore heavily, wake unrefreshed, or have a neck size that makes you a candidate for sleep apnoea, get it assessed; untreated apnoea makes glucose control genuinely harder.

PART THREE

Movement That Actually Happens

You do not need a gym membership, a trainer, or 90 minutes. You need a rule you can keep on a Tuesday when it is raining and you are tired.

24. The fifteen-minute rule

The best-evidenced movement intervention for diabetes is also the cheapest: walk after you eat. A 2022 systematic review and meta-analysis in Sports Medicine pooled seven randomised crossover trials and found that light-intensity walking breaks reduced post-meal glucose by about 17% compared with prolonged sitting, and reduced post-meal insulin as well. A 2023 meta-analysis in the same journal found that exercise after a

meal lowered post-meal glucose excursions significantly more than exercise before a meal, and that the sooner after eating you move, the bigger the effect.

The practical version: ten to fifteen minutes of ordinary walking within half an hour of finishing a meal, three times a day if you can. This is not a workout. It is a stroll. It works because contracting muscle takes up glucose through a pathway that does not need insulin.

> **If you do one thing from this book**
> Walk for ten minutes after your largest meal, every day, starting today. Not after you have bought shoes, not once the weather improves. Today. It is free, it is evidence-backed, and it will show up in your next HbA1c.

25. The weekly plan

WHO and Indian guidance converge on 150-300 minutes of moderate activity a week, plus muscle-strengthening on two or more days. For people with prediabetes who are overweight, RSSDI suggests up to an hour of walking daily; half an hour is the minimum for those at normal weight. Here is a week that fits a working life.

Day	Main session	After meals
Monday	30-minute brisk walk	10 minutes after lunch and dinner
Tuesday	20-minute home strength: squats, wall push-ups, lunges, plank	10 minutes after dinner
Wednesday	30-minute brisk walk	10 minutes after lunch
Thursday	20-minute home strength, same circuit, one more round	10 minutes after dinner
Friday	30-minute brisk walk or 20 minutes of cycling/swimming	10 minutes after lunch and dinner
Saturday	Longer walk, 45-60 minutes, with family	Optional
Sunday	Rest, or gentle yoga and stretching	10 minutes after the heaviest meal

The strength sessions matter more than most people with diabetes realise. Muscle is the largest site of glucose disposal in the body. Losing muscle while losing weight, which happens on any low-calorie diet without resistance work, reduces the very tissue you need to handle carbohydrate. Two short sessions a week protect it.

A home circuit that needs no equipment: 10 squats to a chair, 10 wall push-ups, 10 step-ups on a stair, 20-second plank, 10 glute bridges. Three rounds, resting a minute between rounds. That is 15-20 minutes and it is enough for a beginner.

26. If you sit for a living

Sixty-one percent of Indians are physically inactive by measured activity levels, and desk work is a large part of why. The same meta-analysis that found walking breaks effective also found that simply standing up regularly beats sitting continuously, though walking beats standing.

Three habits that survive a real job: take every phone call standing or walking; set a reminder to stand for two minutes every 45 minutes; and treat the post-lunch walk as a meeting you cannot move. If you commute, get off one stop early, or park at the far end of the lot. These are not optimisations; they are the difference

between 2,000 and 7,000 steps.

27. Yoga, surya namaskar, and what they do

Yoga is genuinely useful for diabetes, but for reasons that are often misdescribed. It improves flexibility, reduces stress and improves sleep, and stress hormones raise glucose. Some studies show modest HbA1c improvements with regular practice. What it does not do is replace the post-meal walk or the strength work; the glucose-lowering effect of a 20-minute asana session is smaller than a brisk walk of the same length.

Use yoga as the thing you will actually do on the days you will not walk, and as the thing that makes the walking possible by improving how you feel. Surya namaskar, done at a pace that raises your heart rate, is legitimate moderate activity. Ten rounds is a real session.

PART FOUR

Measuring Without Obsessing

A number you check once a year is a rumour. A number you check at the right moments is information. The trick is knowing which moments.

28. How often to test what

There are two very different measurements in diabetes care and people confuse them constantly. HbA1c is a three-month average and belongs in a lab. A finger-prick reading is a single moment and belongs in your kitchen. You need both, for different reasons.

Test	How often	Why
HbA1c	Every 3 months until stable, then every 6 months	RSSDI recommends 3-6 months; the ICMR workflow says every 6-12 months, and every 3 months if uncontrolled
Fasting and post-meal glucose	Monthly, or as your doctor advises	Shows whether the plan is working between HbA1c tests
Blood pressure	Every visit, and at home if you own a monitor	Only 48.8% of Indians with diabetes have it controlled
Lipid profile	Once a year, more often after a change in statin	LDL is the target most people never discuss
Urine ACR and serum creatinine/eGFR	Once a year	Detects kidney involvement years before symptoms
Dilated eye examination	Once a year	Retinopathy is silent until it is not
Foot examination	Daily at home, annually by a clinician	The commonest cause of preventable amputation
Dental check	Every 6-12 months	Gum disease worsens glycaemic control; the relationship runs both ways
ECG	Annually, or if you have symptoms	Part of the ICMR annual monitoring set

One test deserves a campaign of its own: the urine albumin-to-creatinine ratio. It is cheap, it is a single urine sample, and it is the earliest signal that diabetes is affecting your kidneys. The ICMR workflow lists it in the

annual monitoring set. In practice, it is skipped in a large share of Indian consultations. Ask for it by name, write it on your file, and do not leave the clinic without a slip.

29. Finger-prick testing that tells you something

Only 16.7% of Indians with diabetes monitor at home, and among those on insulin only 36.9% do. Yet self-monitoring is what turns a vague sense of 'doing okay' into a decision about dinner.

You do not need to test seven times a day forever. You need to test strategically for two weeks, learn your patterns, and then test at the moments that matter.

Moment	What it tells you	Target
On waking, before food or tea	Your fasting glucose, driven by overnight liver output	80-130 mg/dL
Two hours after the first bite of a meal	How that specific meal treated you	Below 180 mg/dL
Before bed	Whether you are safe overnight, especially on insulin	100-140 mg/dL is a common working range
Any time you feel shaky, sweaty, confused or very hungry	Whether you are hypoglycaemic	Below 70 mg/dL needs action
Before driving	Safety, if you take insulin or a sulfonylurea	Above 90 mg/dL
During illness or fever	Sick days push glucose up sharply	Every 4 hours

The most valuable single habit is the paired test: measure before a meal and two hours after it, for the same meal. That pair tells you what that food did to you, personally, which no book and no dietician's list can tell you. Do this for your five most common meals over two weeks and you will have a personal food map more accurate than any glycaemic index table.

Practical points: wash and dry your hands, because fruit juice or sugar on a fingertip produces a falsely high reading. Use the side of the fingertip, not the pad. Rotate fingers. Check that your strips are in date and the bottle is closed; strips degrade in Indian humidity. And accept that meters are allowed a margin of error, so do not make decisions on a difference of 5 mg/dL.

30. Continuous glucose monitoring, honestly

A CGM is a small sensor worn on the arm that reads glucose every few minutes and sends it to your phone. It is genuinely transformative for some people and a distraction for others.

Where it earns its cost: if you take insulin and need to see overnight trends and prevent lows; if your HbA1c does not match your finger-prick readings; if you want to learn which foods spike you without twenty finger pricks a day; if you have frequent unexplained highs or lows. The ICMR workflow lists CGM alongside self-monitoring as a monitoring option.

Where it does not: if you will obsess over every fluctuation. Normal glucose is not a flat line; it rises after every meal and dips with exercise, and a sensor makes that visible in a way that can create anxiety where none existed. Set the alarms wide, look at the daily pattern rather than the last reading, and use a two-week sensor as a learning tool rather than a permanent attachment.

Cost is the practical constraint, and sensor prices change frequently, so check current pricing with your pharmacy or diabetologist rather than relying on any figure in print. If cost is a barrier, the paired finger-prick

method described earlier gives you most of the insight for a fraction of the price.

31. Reading the pattern, not the number

Three patterns account for most unexplained glucose problems. Learn to recognise them and you can often solve them yourself before your next appointment.

Pattern	What it looks like	What it usually means
High fasting, decent daytime	Waking glucose 150+, but readings after meals are reasonable	Overnight liver glucose release, often from a late or large dinner, poor sleep, or too little medication at night. Sometimes the dawn phenomenon.
The post-meal cliff	Good before eating, 220+ two hours after	The meal itself, or its portion, or the order you ate it in. This is the most fixable pattern in the book.
The afternoon dip	Shaky, hungry, sweaty around 3-5 pm	Too large a lunch spike followed by a reactive fall, or medication peaking. Test it rather than guessing.
Unexplained highs for days	Everything is up, nothing changed	Infection, dental problem, steroid medication, stress, or a bad batch of strips. Rule out illness and check your meter.

The single most common self-inflicted error is changing three things at once. Change one variable, hold it for a week, measure. That is the difference between a plan and a guess.

32. The twenty-minute appointment

An Indian diabetes consultation is often six minutes. You can make six minutes work, but only if you arrive prepared. Bring these five things, every time.

1. Your last three HbA1c values with dates, written down. Not in an app nobody opens.
2. Your home readings for the past two weeks, with the meal they followed.
3. Your current medicine list with doses, including anything you bought over the counter.
4. Any new symptom: numbness, blurring, foot sores, chest discomfort on walking, swelling, unexplained weight loss.
5. Three specific questions. Written. Because you will forget them the moment the doctor says 'next'.

Questions worth asking, in the order that gets answers:

- Are all three of my numbers at target, or only my sugar? What is my LDL and my blood pressure?
- When was my last urine ACR and eye check? Should we do them now?
- Is my current medicine the right one for my weight and my heart risk, or just the cheapest?
- What is my personal target, given my age and other conditions?
- If my readings are good for three months, can we discuss reducing anything?

33. Travelling with diabetes

Travel is where routines break, and where a small amount of preparation prevents the problems that ruin a trip.

- Carry twice the medicine you need, split across two bags. A delayed flight is a common reason for an unexpected gap in supply.
- Keep all medicines, insulin and strips in your hand baggage, never in checked luggage. Insulin can freeze in the cargo hold, and frozen insulin is dead.
- Carry a copy of your prescription and a short letter from your doctor listing your medicines. It helps at airport security, which permits insulin and syringes with documentation, and it helps if you need a pharmacy in an unfamiliar city.
- Insulin in use should stay cool, not cold. A simple insulin pouch or an insulated bag with a cool pack works. Never leave it in a parked car or a bike compartment in Indian summer.
- Test strips degrade in heat and humidity. Keep them in a sealed box with the desiccant, and do not store them in the car or the bathroom.
- On trains, order the regular thali rather than the fried snacks, and carry roasted chana, nuts, fruit and a bottle of water. The pantry car is not where good decisions happen.
- Walk at every long station halt. Ten minutes of walking after a train meal is the same intervention as at home.
- Carry glucose tablets, always. Travel delays meals, and delays cause lows if you are on insulin or a sulfonylurea.
- If you cross time zones on insulin, get a written plan from your doctor before you fly. Do not improvise a schedule on the plane.
- Expect higher readings for a day or two. Travel is stress, disrupted sleep and unfamiliar food. One high reading is not a trend.
- Never walk barefoot, including in a hotel room or on a beach. This is the trip where a small foot injury becomes a serious one.

PART FIVE

Medicines, Without the Noise

This part is information, not prescription. Nothing here tells you what to take. It tells you what exists, what it does, what it costs, and what to ask - so that the decision is a conversation rather than a guess.

34. The ladder, and why it is climbed earlier now

Indian guidelines have moved away from the old 'start with metformin and wait' approach for people who present with high sugars. The ICMR Standard Treatment Workflow is specific: for an HbA1c below 8.5%, monotherapy with metformin is the usual starting point; for an HbA1c between 8.5% and 10%, dual therapy from the outset is recommended, pairing metformin with a sulfonylurea, a thiazolidinedione, a DPP-4 inhibitor, an SGLT2 inhibitor, an alpha-glucosidase inhibitor or a GLP-1 receptor agonist.

That is a meaningful shift in philosophy. It says: if you are far from target, do not spend a year finding out that one drug was not enough. Start with two, from different classes, and get there faster.

Class	How it works	What to know
Metformin	Reduces glucose production by the liver and improves insulin sensitivity	First-line for most people. Take with food. Gut upset is common in the first weeks and usually settles. Long-term use can lower vitamin B12; ask about periodic testing.

Class	How it works	What to know
Sulfonylureas (glimepiride, gliclazide)	Make the pancreas release more insulin	Effective and very cheap. Carry a real risk of hypoglycaemia and of weight gain. Do not skip meals on these.
DPP-4 inhibitors (sitagliptin, linagliptin, teneligliptin)	Prolong the action of your own GLP-1	Well tolerated, weight-neutral, low hypoglycaemia risk. Modest glucose-lowering power. Very widely prescribed in India.
SGLT2 inhibitors (dapagliflozin, empagliflozin)	Make the kidney excrete glucose in urine	Also protect heart and kidney in the right patients. Increase urination and genital fungal infection risk; keep hydration up. Important: they can rarely cause a serious condition called euglycaemic ketoacidosis, so stop them before surgery or during severe illness as your doctor directs.
GLP-1 receptor agonists (semaglutide, liraglutide, dulaglutide)	Mimic a gut hormone that increases fullness and glucose-dependent insulin release	Substantial weight loss and cardiovascular benefit in many patients. Nausea and constipation early on. See the semaglutide price section for the 2026 change.
Insulin	Replaces or supplements your own insulin	Needed when oral therapy is insufficient, in significant weight loss, in very high glucose with symptoms, in pregnancy, and in type 1. Not a failure. See the insulin section.

Two things worth knowing about the Indian market. First, combination pills are common and convenient, which improves adherence but makes it harder to adjust one component. Second, brand matters less than molecule: the same salt from a different company is legally the same medicine, and in India generics are how almost everyone is treated. What matters is that you take it consistently and that someone reviews it.

35. The question of 'newer' drugs

Every few months a patient asks whether they should be on the newest medicine they saw on television. The honest answer is that 'newer' is not the same as 'better for you'. Two questions decide it.

First, do you have heart or kidney disease, or high risk of either? If so, SGLT2 inhibitors and GLP-1 receptor agonists have outcome evidence beyond glucose lowering, and guidelines increasingly position them for exactly those patients regardless of HbA1c. Second, is weight a major part of your problem? If yes, the GLP-1 class and metformin are more attractive than sulfonylureas, which tend to add kilos.

Conversely, if your HbA1c is 7.4%, you have no heart or kidney disease, you tolerate metformin, and cost is a real constraint, an expensive addition buys you very little. The cheapest effective regimen taken consistently beats an expensive regimen taken sporadically, and this is not a platitude; adherence is the dominant variable in real-world Indian outcomes.

36. The 2026 semaglutide price change, explained

This is the most significant practical development in Indian diabetes care in years, and most people have heard a confused version of it.

Semaglutide, sold by Novo Nordisk as Ozempic for diabetes and Wegovy for weight management, was protected by a patent in India that expired in March 2026. Within a day, at least seven major Indian manufacturers launched generic versions, priced between 50% and 80% below the innovator product. Novo Nordisk then cut prices on its own brands from 1 April 2026: Ozempic by 36% and Wegovy by 48%.

Product	Approximate monthly cost after the 2026 changes
Sun Pharma generic semaglutide (diabetes indication)	Rs 750-1,300 per month
Natco generic semaglutide (vial)	Around Rs 1,250 per month
Alkem generic (pre-filled pen)	From around Rs 1,800 per month
Zydus generic (injectable and reusable device)	Around Rs 2,200 per month on average
Dr Reddy's generic (diabetes, pen)	Around Rs 4,200 per month
Torrent oral semaglutide tablet	Around Rs 149 per tablet
Innovator Ozempic (Novo Nordisk)	Rs 2,200-2,793 per week
Innovator Wegovy (Novo Nordisk)	Rs 2,700-4,100 per week
Oral semaglutide for diabetes (Rybelsus)	Over Rs 8,000 per month at last reported pricing

Read this carefully, because the numbers carry traps. The costs above are not like-for-like: some are for the diabetes indication and some for obesity; some are vials requiring syringes, some are pens; and dose escalation means your monthly cost rises as your dose rises. Prices are also moving fast and vary by pharmacy and city. Treat this table as evidence that the class has become dramatically more affordable, not as a price list.

What has not changed: these are prescription medicines. They cause nausea, vomiting and constipation in a significant minority, they require dose titration, they are not recommended in pregnancy, and they need monitoring. The oral Wegovy tablet, despite being approved in the UK and EU, had not received Indian marketing approval as of late 2026; the Indian regulator asked the company for India-specific bioavailability and trial data at its September 2026 meeting. So if someone offers you 'the Wegovy tablet' in India, ask what it actually is.

> **The right question to ask your doctor in 2026**
> Not 'can I get Ozempic?' but 'given my HbA1c, my weight, my heart and kidney risk, and my budget of about Rs X per month, is a GLP-1 the best use of that money compared with a statin, better blood pressure control, or a consistent metformin routine?' That question gets a useful answer. The first one gets a prescription or a shrug.

37. Insulin, if and when

Insulin is the most effective glucose-lowering medicine that exists, and in India it carries a stigma that costs lives. Three facts to dismantle it.

First, needing insulin is not a punishment for bad behaviour. Type 2 diabetes is progressive; beta-cell function declines over years in most people, and at some point the pancreas genuinely cannot keep up. Second, starting insulin earlier and briefly, to bring very high glucose down, can actually rest the pancreas and improve control afterwards. Third, in pregnancy, in severe infection, after surgery, and in significant unintentional weight loss, insulin is the correct and safest choice.

Practical matters that nobody explains properly. Storage: insulin in use can be kept at room temperature, but in an Indian summer above roughly 30 degrees Celsius it degrades, so keep it cool and never in a car or a bike compartment. Never freeze it. Injection technique: rotate sites within the same general area, abdomen or thigh, and do not reuse needles to the point of bending; a blunt needle hurts more and delivers less predictably. And if you take insulin, self-monitoring is not optional, it is part of the treatment.

38. Hypoglycaemia: the thing to get right

A low is defined as glucose below 70 mg/dL. It happens most with insulin and sulfonylureas, and it is the main reason doctors sometimes accept a higher HbA1c than ideal.

Stage	What you feel	What to do
Mild	Shakiness, sweating, sudden hunger, palpitations, anxiety	15 g of fast carbohydrate, wait 15 minutes, retest. Repeat if still below 70.
Moderate	Confusion, difficulty concentrating, slurred speech, blurred vision	Same, but do not leave the person alone. Someone should stay with them.
Severe	Unconsciousness, seizure, inability to swallow	Nothing by mouth. This is a medical emergency. Call for help immediately and get emergency care.

Fifteen grams of fast carbohydrate is: three teaspoons of sugar or glucose powder in water, or 150 ml of fruit juice or regular soft drink (not diet), or three to four glucose tablets. Then, once you are above 70 mg/dL, eat a small snack with protein and carbohydrate, such as a slice of bread with curd, so you do not dip again.

The 15-15 rule is the core of it: 15 grams, 15 minutes, retest. Do not treat a low with a full meal, and do not treat it with chocolate, which is high in fat and absorbs too slowly to help quickly.

Prevention beats treatment. Do not skip meals on sulfonylureas or insulin. Test before driving and do not drive below 90 mg/dL. Carry glucose always, in your bag, your car, your desk. Tell the people you live with what to look for. And if you have had a severe low, that is a conversation with your doctor within days, not months, because it usually means a dose or a regimen needs changing.

39. Sick days, fever, surgery and dental work

Illness raises glucose sharply, even when you are eating less. This is the counterintuitive fact that catches people out: fever, infection, vomiting and diarrhoea all push glucose up, while appetite falls.

- Keep taking your diabetes medicines unless a doctor tells you to stop, even if you are eating little. The exception is SGLT2 inhibitors during severe illness or before surgery, which your doctor will direct.
- Test every four hours during significant illness, and more often if you are on insulin.
- Drink water steadily. Dehydration makes everything worse and is the commonest reason a sick day becomes a hospital admission.
- If you cannot keep food down, use fluids with some carbohydrate: buttermilk, soup with a little rice, diluted juice, or ORS if there is diarrhoea.
- Know the emergency signs: vomiting that will not stop, deep rapid breathing, confusion, severe abdominal pain, glucose persistently above 250-300 mg/dL, or ketones if you have been told to test for them. These mean hospital, not home remedies.
- Before any surgery, tell the surgeon and the anaesthetist that you have diabetes, and expect a plan for holding or adjusting medicines. Do not decide this yourself.
- Dental infections are a real and underrated cause of unexplained high glucose. A persistent tooth or gum problem can hold your HbA1c up on its own.

40. Night shifts, odd hours and long duty

India runs on shift work: BPO and IT support, nursing, factory floors, transport, security, hospitality, and the small shop that opens at six. Shift work is genuinely harder for glucose control, and pretending otherwise is

unhelpful. What helps is a plan built for the shift rather than a plan borrowed from a nine-to-five.

- Eat your main meal before the shift begins, not at 2 am. A large meal in the middle of the night produces a bigger and longer glucose rise than the same meal eaten in the evening.
- Keep the 2-3 am meal small and protein-led: curd, eggs, paneer, a bowl of dal, some nuts. Not a full thali, and not the canteen's fried snacks.
- Stop caffeine about six hours before you plan to sleep. A 6 am coffee when you finish at 7 am is why you sleep badly at 9 am.
- Protect the sleep block. Dark room, phone on silent, family told not to disturb. Short sleep raises insulin resistance the next day, which is exactly the day you are working.
- Test at consistent points in your shift rather than at consistent clock times, so the readings are comparable.
- If you take a sulfonylurea or insulin, medicine timing across a rotating shift needs a written plan from your doctor. This is one of the few situations where guessing is genuinely unsafe.
- Walk on your break. Ten minutes outside at 3 am does more for your glucose than a second cup of tea.
- Do not skip meals to compensate for a night of eating badly. The resulting low, or the resulting binge, is worse than the meal you skipped.

41. Supplements, powders and 'cures'

India's diabetes supplement market is enormous and almost entirely unregulated in practice. Here is a filter you can apply to any bottle.

Things with some evidence, in specific contexts: vitamin B12 if you have been on metformin for years, since it can deplete B12; vitamin D if you are deficient, which is common; and fibre supplementation, which has modest effects on glucose and satiety. All of these are corrections of a deficiency or a modest add-on, not treatments.

Things with weak or absent evidence for reversing diabetes: bitter gourd (karela) juice, fenugreek (methi) seeds in large doses, cinnamon capsules, jamun seed powder, chromium picolinate, alpha-lipoic acid for glucose control (though there is some evidence for nerve pain), and the long list of 'diabetes cure' powders sold with a WhatsApp number. None of them will harm you much, and none of them will do what a 5% weight loss and a daily walk will do.

Things that genuinely harm: anything that claims you can stop your medicine. Anything sold as a 'course' with a monthly fee and a promise of reversal in 90 days. Anything containing undisclosed steroids, which some 'ayurvedic' joint and pain products in India have been found to contain. And high-dose supplements in kidney disease, where potassium and phosphorus load matters.

If you want to try something complementary, tell your doctor what it is, keep taking your prescribed medicine, and give it eight weeks with an HbA1c or paired readings before and after. That is the test any honest product should pass.

42. Twenty questions people actually ask

These are the questions that come up in almost every conversation about diabetes in India. The answers are short on purpose.

Can I eat rice at all?

Yes. Rice is not poison, and banning it is why so many plans fail. Eat it in a smaller quantity, after the vegetables and protein, and walk afterwards. Half a katori of rice with a proper dal and sabzi behaves very differently from two katoris of rice with a pickle.

Is banana allowed?

One medium banana is about 25-30 g of carbohydrate, roughly the same as one and a half rotis. Eat it whole, not as a shake or juice, ideally with a few nuts, and count it as part of the meal's carbohydrate rather than as a free extra. A small banana is a better choice than a large one.

Will I have to take medicine forever?

Possibly, and possibly not. Type 2 diabetes is progressive for most people, so needing more over time is normal biology, not personal failure. But some people who lose meaningful weight and keep it off can reduce or stop medication under supervision, and some reach remission. Nobody can promise which group you are in.

Is honey or jaggery better than sugar?

No. Both raise glucose similarly. Jaggery carries trace minerals, which is not a reason to eat more of it. If you use either, count it as sugar and keep the total under 25-30 g a day.

Does karela or methi actually work?

There is some laboratory and small-study interest in both, and no reliable evidence that either reverses diabetes. They will not harm you in food quantities. They will not do what a 5% weight loss and a daily walk do. If a product claims otherwise, see the last section of this book.

Is brown rice or millet actually better?

Slightly, in fibre and micronutrients, and worth eating for that reason. But the 2025 Nature Medicine analysis found that swapping refined cereals for whole wheat or millet flour without reducing total carbohydrate did not lower diabetes risk. Quality matters, quantity matters more.

Do I need to test my sugar every day?

Not necessarily, if you are stable and on metformin alone. You do need to test strategically: paired before-and-after readings for two weeks to learn your food patterns, and then at the moments that matter. If you are on insulin or a sulfonylurea, regular testing is part of the treatment, not optional.

Can diabetes be cured?

No, but it can go into remission, defined as an HbA1c below 6.5% for at least three months without glucose-lowering medication. Remission is real and worth pursuing. It is also not the same as a cure, because the underlying tendency remains and weight regain usually brings the glucose back.

Is ghee bad for me?

Ghee is saturated fat and it counts, but a teaspoon for flavour is not the problem. The problems are quantity, repeated deep-frying, and vanaspati or bakery shortenings containing trans fats. Measure the oil you pour; most Indian households use two to three times the recommended 27 g a day.

Will I end up on insulin?

Many people with type 2 diabetes eventually need it, because insulin production declines over years. Starting it is not a punishment and not the end. It is the most effective glucose-lowering treatment available, and in pregnancy, severe infection or very high glucose it is the safest choice.

Is diabetes hereditary?

Strongly, yes. A first-degree relative with diabetes is one of the recognised reasons to get tested, and Indian guidelines treat it as a risk factor in its own right. That is an argument for screening your siblings and children early, not for fatalism: genetics loads the gun, lifestyle pulls the trigger.

Are sugar-free and diabetic products safe?

Safer than the sugar versions, and usually not healthy. A diabetic biscuit is still refined flour and fat. Sugar alcohols can cause bloating and loose motions in quantity. Read the back of the pack, not the front.

Can I donate blood?

Usually yes, if your glucose is well controlled and you have no complications, but eligibility rules vary by blood bank and by medication. Ask the blood bank directly and mention your medicines. Do not decide this from a general rule.

Should I take vitamin B12?

Possibly, if you have been on metformin for years, because long-term use can lower B12 levels. Get it tested rather than guessing, and treat a deficiency rather than taking a supplement as insurance.

Does stress really affect my sugar?

Yes, measurably. Stress hormones raise glucose, and poor sleep from stress raises insulin resistance the next day. Anyone who tells you it is 'all in your head' is describing the mechanism and calling it a dismissal. Sleep, walking and breathing exercises are legitimate treatment, not luxuries.

Is fruit juice as good as fruit?

No. Juice removes the fibre and the chewing, and a glass can carry the sugar of three fruits in thirty seconds. Eat fruit whole. The official guidance is explicit about preferring fresh fruit and avoiding juices.

Can I drink alcohol?

If you drink, occasionally and with food. Alcohol lowers glucose acutely and can cause a delayed low overnight, especially with insulin or sulfonylureas. Never on an empty stomach, and check your glucose before bed on a night you have drunk.

Do I need to see a specialist or is my family doctor enough?

A good family physician manages most type 2 diabetes well. See a diabetologist or endocrinologist if your HbA1c stays above target on two or three medicines, if you have kidney or heart disease, if you are on insulin and struggling, or if you are planning a pregnancy.

How often should I get an HbA1c?

Every three months until you are at target and stable, then every six months. Indian guidelines differ slightly, allowing six to twelve months in stable patients, but if anything has changed, three months is the useful interval.

What is the single most important thing I can do this week?

Walk for ten minutes after your largest meal, every day for seven days. It is free, it is the best-evidenced movement intervention in diabetes, and it will be visible in your next HbA1c. Everything else in this book is easier once that habit exists.

PART SIX

Protecting the Organs That Pay for It

Glucose is the number you watch. Complications are what actually happen to you. Almost all of them are preventable or delayable, and almost all are found late because nobody looked.

43. The annual screening calendar

The ICMR Standard Treatment Workflow sets out a clear annual set. Put these on your phone calendar as repeating reminders with the month attached, because 'I will do it later' is how retinopathy is discovered at the bleeding stage.

Check	Frequency	What it catches early
Dilated retinal examination by an ophthalmologist	Yearly	Retinopathy, a leading cause of preventable vision loss in working-age adults
Urine albumin-to-creatinine ratio	Yearly	Kidney damage, years before creatinine rises
Serum creatinine and eGFR	Yearly	Kidney function trend
Foot examination including monofilament and vibration sensation	Yearly by a clinician, daily by you	Loss of protective sensation, the pathway to amputation
ECG	Yearly	Silent heart disease; diabetes blunts the usual warning symptoms
Lipid profile	Yearly	LDL, the target most often missed
Blood pressure	Every visit	The largest single driver of kidney and stroke risk
Dental examination	Every 6-12 months	Gum disease, which worsens control
Vaccinations	As advised	Influenza and pneumococcal disease are more dangerous with diabetes

44. Eyes and kidneys: the two that go quiet

Diabetic retinopathy has no symptoms until it affects the macula or bleeds. By the time you notice blurring, treatable early changes may already have been missed. An annual dilated fundus examination is one of the highest-value twenty minutes in all of medicine, and it costs a few hundred rupees at most district hospitals. If you have not had one in the last year, that is this week's task, not this year's.

Kidney disease follows a similar pattern. The first sign is a small amount of albumin in the urine, detectable by the ACR test, at a stage when tight glucose and blood pressure control and specific medication can slow progression substantially. The ICMR workflow targets a blood pressure of 130/80 in chronic kidney disease, tighter than the general target. If your ACR is raised, that is a conversation about blood pressure and about SGLT2 inhibitors, both of which protect kidneys.

45. Your feet, and the two minutes that save them

Amputation is the complication people fear most, and almost every case begins the same way: a blister, a corn or a small cut that went unnoticed because sensation was reduced. The pathway is predictable, which

means it is also interruptible.

The daily two minutes: after bathing, sit down, and look at the sole, the heel, between the toes, and the top of each foot. Use a mirror for the sole, or ask someone. Look for cuts, blisters, redness, swelling, cracks, or colour change. Feel for heat. If anything is new and not healing in 48 hours, see a doctor.

- Never walk barefoot, indoors or outdoors. Not to the temple, not to the kitchen, not on hot marble in summer.
- Test bath water with your elbow or a thermometer, not your foot. Neuropathy means you can scald yourself without feeling it.
- Cut nails straight across, not into the corners, and never cut corns or calluses yourself. See a podiatrist or a trained clinician.
- Wear shoes that fit, break new shoes in slowly, and check inside the shoe with your hand before putting it on, for a stone, a staple, or a fold in the insole.
- Moisturise dry skin but not between the toes, where moisture causes fungal infection.
- Stop smoking. It is the single largest amplifier of peripheral artery disease in diabetes.
- Any wound that is red, warm, swollen, or has discharge is urgent. Do not wait a week to see if it settles.

46. Heart, brain, nerves and teeth

Cardiovascular disease is what most people with diabetes eventually die of, and the risk begins well before any symptoms. Three things reduce it, in order of impact: blood pressure control, LDL control, and not smoking. Glucose control matters, but it is not the largest lever for the heart.

Silent heart attacks are a real phenomenon in diabetes, because neuropathy dulls the pain. Do not dismiss breathlessness on climbing stairs, unusual fatigue, or discomfort in the jaw, neck or upper abdomen. Get an ECG.

Nerve damage, peripheral neuropathy, usually starts in the toes as numbness, burning or a loss of the ability to feel a light touch. It is why foot care is not optional. If you have burning feet at night, tell your doctor; there are treatments for the pain even though the nerve damage itself is not reversible.

Gum disease deserves more attention than it gets in India. Periodontal infection worsens glycaemic control and glycaemic control worsens periodontal disease, a genuine two-way street. Brushing twice, cleaning between teeth, and a professional cleaning every six to twelve months is part of diabetes care, not a separate matter.

47. Complications at a glance

Eight things diabetes does to the body, what the first sign usually is, how it gets caught, and what actually slows it down. Almost every row is preventable or delayable, and almost every row is caught late because nobody looked.

Complication	First sign	How it is caught	What slows it
Retinopathy (eyes)	Often none at all; later, blurred or fluctuating vision	Annual dilated eye examination	Glucose and blood pressure control; laser and injection treatment when needed

Complication	First sign	How it is caught	What slows it
Kidney disease	None; later, swelling, fatigue	Annual urine ACR plus creatinine and eGFR	Blood pressure below 130/80, glucose control, SGLT2 inhibitors in the right patients, less salt and protein moderation if advised
Peripheral neuropathy	Numbness, burning or tingling in the toes, worse at night	Annual foot examination with monofilament and vibration testing	Glucose control; foot protection; specific treatment for nerve pain
Foot ulcers and amputation	A blister, corn or cut that does not heal	Your daily two-minute check, and a clinician's annual exam	Never barefoot, correct footwear, early treatment of any wound
Heart attack and stroke	Breathlessness on exertion, unusual fatigue; sometimes nothing at all	Annual ECG, blood pressure and lipids every visit	Blood pressure, LDL, not smoking - in that order
Peripheral artery disease	Cramping calf pain on walking, cold feet, slow-healing sores	Foot pulses on examination, Doppler if suspected	Stop smoking; walking; lipid and glucose control
Gum disease	Bleeding gums, persistent bad breath, loose teeth	Dental examination every 6-12 months	Brushing twice, cleaning between teeth, professional cleaning, glucose control
Infections, including fungal	Repeated thrush, boils, urinary infections, slow-healing skin	Clinical examination; urine tests if recurrent	Glucose control, hydration, hygiene, avoiding tight synthetic footwear

Two things are worth noticing about this table. First, four of the eight have no symptoms until they are advanced, which is why the annual screening set exists at all. Second, the interventions in the right-hand column are almost all cheap, and none of them are supplements.

PART SEVEN

Money, Insurance and Family

Diabetes is a lifelong cost, and the Indian system has more free and cheap support than most people use. This part is about paying less and being covered when it matters.

48. What the government already gives you, free

Most people with diabetes in India do not know what they are entitled to. This is the list worth memorising.

Scheme or service	What you get
Ayushman Arogya Mandir (the renamed health and wellness centres)	Free screening for diabetes and hypertension for everyone aged 30 and above, free essential medicines and diagnostics, counselling, and teleconsultation through eSanjeevani. There were about 1.87 lakh functional centres as of November 2026.
NP-NCD, the national programme for non-communicable diseases	Free medicines for diabetes and hypertension at AAMs, PHCs, CHCs and district hospitals, with monthly follow-up recorded on the national NCD portal. Between January and June 2025, 1.11 crore people were under treatment for hypertension and 64.1 lakh for diabetes through this route.

Scheme or service	What you get
District NCD clinics and CHC NCD clinics	770 district NCD clinics, 6,410 CHC-level NCD clinics and 233 cardiac care units, with counsellors for lifestyle advice.
PM-JAY (Ayushman Bharat)	Health cover of Rs 5 lakh per family per year for secondary and tertiary care, for eligible families, covering hospitalisation including diabetes-related complications and procedures.
eSanjeevani	Free government teleconsultation with a doctor from home, useful for follow-ups and prescription renewal where you cannot travel.

The practical implication is significant. If cost is the barrier between you and regular follow-up, you can get screening, medicines and monthly review free at a government facility, and pay only for the occasional lab test. That is a far better outcome than skipping medicines. Ask at your nearest Ayushman Arogya Mandir for the NCD screening and the free drug list; the Community Health Officer there manages exactly this.

49. Insurance when you already have diabetes

The most common misconception in India is that diabetes makes you uninsurable. It does not. It changes the terms, and the terms changed substantially in your favour in 2024.

Rule	What it means for you
Pre-existing disease waiting period: maximum 36 months	Reduced from 48 months by IRDAI in 2024, applicable to new policies and renewals after 30 September 2024. Diabetes-related claims are covered after this waiting period.
Moratorium period: 60 months	Reduced from 96 months. After five years of continuous coverage, no policy or claim can be contested on grounds of non-disclosure or misrepresentation, except for established fraud.
Portability carries your credits	If you switch insurers, the waiting period you have already served, and the moratorium credit, transfer with you. You do not restart the clock.
Renewal cannot be refused because you claimed	An insurer cannot deny renewal on the ground that you made a claim in a previous year, and cannot re-underwrite you unless you increase the sum insured.
Products must be offered to all ages and to people with pre-existing conditions	IRDAI's 2024 master circular requires insurers to make products available covering all ages and existing medical conditions, including chronic conditions. Pricing and underwriting remain the insurer's call.
Grace period of 15 or 30 days	Depending on your premium payment frequency, and credits are protected if you renew within it.

Three practical rules. First, disclose. Diabetes is a pre-existing condition and hiding it is the single commonest reason claims are rejected; after the moratorium you are protected, but only if you got there honestly. Second, buy earlier rather than later, because the waiting period runs from inception and every year you delay is a year of no cover. Third, check whether your policy has a diabetes-specific sub-limit or a co-payment on pre-existing conditions, and read the Customer Information Sheet, which insurers are required to provide in simple language covering waiting periods, sub-limits and exclusions.

50. What this actually costs, monthly

Most families never add up the real monthly cost of diabetes, and then are surprised when the annual figure appears. Fill this in once, honestly, and you will know where the money goes.

Item	Typical monthly spend (fill in yours)	Notes
Medicines	Rs _____	Ask for generic names; the same molecule is often far cheaper
Test strips	Rs _____	The largest hidden cost for people who monitor properly
Lab tests, averaged monthly	Rs _____	HbA1c 2-4 times a year, lipids and ACR once
Doctor consultations	Rs _____	AAM and PHC visits are free
Travel to appointments	Rs _____	eSanjeevani can remove most of this
Supplements and 'diabetes products'	Rs _____	The line most worth cutting to zero
Insurance premium, averaged monthly	Rs _____	Rises with age; buy early

Two ways to cut this without cutting care. Switch to generic names and ask the pharmacist for the cheapest brand of the same salt; the difference between brands of metformin can be several hundred rupees a month for identical medicine. And move routine follow-up and free medicines to a government facility while keeping the private specialist for the annual review. Many people run this hybrid and halve their spend.

51. Family: the part nobody plans for

Type 2 diabetes is strongly heritable. If a parent has it, your risk is materially higher, and the ICMR screening criteria list a first-degree relative with diabetes as a reason to test. If you have it, your siblings and children should be screened, and the cheapest moment to catch it is now.

Household change beats individual willpower. If one person in the house is on a plate rule and everyone else is eating four rotis and a sweet every night, the plan fails within a month. The most effective intervention in Indian households is a kitchen-level change: less rice cooked in the same pot, one extra dal, a bowl of curd on the table, no biscuits in the cupboard. Nobody has to be on a diet; the whole family simply eats the same better-proportioned food.

On pregnancy, three facts worth knowing. All pregnant women should be screened for gestational diabetes, with a 75 g oral glucose tolerance test at 24-28 weeks if the first-trimester screen is normal. In pregnancy, a fasting glucose between 92 and 125 mg/dL is treated as gestational diabetes rather than as pre-existing diabetes. And women who have had gestational diabetes should be tested at six weeks after delivery and then at least every three years, because their lifetime risk of type 2 diabetes is high. The ICMR workflow also advises that people with prediabetes be tested yearly.

For children, the advice is unglamorous and correct: do not put a child on a restrictive diet. Reduce sugary drinks to zero, keep fruit juice out of the house, make one meal a day a real cooked meal, and let them play outside for an hour. Screen children who are overweight and have a family history, particularly around puberty.

PART EIGHT

The Worksheets

Print these pages, or copy them into a notebook. A plan you cannot see is a plan you will not follow.

52. Your first thirty days

Four weeks, one change per week, in the order that produces the fastest visible result. Do not skip ahead; each week's habit is the foundation for the next.

Week	The one change	How you know it worked
1	Walk 10 minutes after your largest meal, every day	Seven days ticked on the calendar
2	Rebuild the plate: half vegetables, a quarter protein, a quarter cereal. Same food, new proportions	You can point at the protein on your plate at every meal
3	Fix breakfast: halve the cereal, add a protein	Your 2-hour post-breakfast reading drops
4	Add protein to lunch and dinner, and cut cereal by one unit per meal	Fewer 3 pm cravings; a lower post-dinner reading

Then measure. Get an HbA1c at the end of the month only if it has been at least three months since the last one; otherwise use paired finger-prick readings as your evidence. If nothing has moved after four weeks of genuine adherence, the problem is not your willpower, it is the plan, and it needs a doctor's review.

53. The weekly meal planner

Fill this in on a Sunday. The point is not the perfect week; it is that 5 pm on Wednesday does not require a decision.

	Breakfast	Lunch	Snack	Dinner
Mon				
Tue				
Wed				
Thu				
Fri				
Sat				
Sun				

Rules for filling it: every breakfast includes a protein; every lunch and dinner has a visible protein and a vegetable; one meal a week is a free meal, planned in advance rather than fallen into; and the snack column is filled with things that are already in the house.

54. Glucose log

Two weeks of paired readings will teach you more than a year of vague impressions. Test before and two hours after the same meal, note what you ate, and look for the pattern.

Date	Before	After 2h	What you ate	Walked?

Targets to compare against: before a meal 80-130 mg/dL, two hours after the first bite below 180 mg/dL. A single reading above target is information, not failure. Three of the same reading after the same meal is a pattern, and a pattern is something you can change.

55. The card in your wallet

Fill this in and keep it in your wallet or phone case. It takes two minutes and it matters most on the day you cannot speak for yourself.

Field	Yours
Name and age	
I have (type of diabetes, year of diagnosis)	
My medicines and doses	
I am allergic to	
My doctor and phone number	
Emergency contact and phone number	
If I am confused or unconscious, this may be a low blood sugar. Give sugar if I can swallow.	Yes
Nearest hospital	

56. How to test any diabetes claim you read

You will be sold something this month. A powder, a course, a five-day retreat, a device, a diet plan with a monthly fee. Eight questions will tell you within two minutes whether it deserves your money.

1. Who is making the claim, and are they selling the thing they are claiming? A trial funded by a company that profits from the result is weaker evidence, not worthless, but weaker.
2. Is there a human trial, or only a laboratory finding, an animal study, or a testimonial? Karela lowering glucose in a petri dish tells you almost nothing about your dinner.
3. How many people, and for how long? Twelve people for four weeks is a signal to investigate, not a result to act on.
4. Compared to what? A supplement that lowers HbA1c by 0.2% sounds impressive until you learn the control group got nothing and a daily walk lowers it by far more.
5. What is the absolute effect? '28% risk reduction' can mean six people treated to prevent one case, which is worth it, or a thousand treated to prevent one, which may not be.

6. Does it conflict with the Indian guidelines? If a product promises what ICMR, RSSDI and the national programme say is not possible, the product is wrong, not the guidelines.
7. What is the harm and the cost? Every rupee spent on a useless powder is a rupee not spent on test strips, a statin, or a pair of walking shoes.
8. What happens if you do nothing for eight weeks? Often the honest answer is: nothing bad, and you saved the money.

A worked example, on the claim you will hear most
'Switch to millet atta and your diabetes will improve.' Test it. Who: sellers of millet flour, and well-meaning clinicians. Evidence: the 2025 Nature Medicine analysis of 18,090 Indian adults tested exactly this substitution and found no reduction in diabetes risk (odds ratio 0.94, confidence interval 0.57 to 1.56). Absolute effect: none detectable. Harm: none, millets are good food. Cost: a premium over wheat flour. Verdict: eat millets because they are nutritious and the national guidelines recommend them at 30-40% of your cereal intake, not because they will fix your glucose on their own. The thing that fixes your glucose is eating less cereal and more protein.

57. Sources and how this was verified

Every number in this playbook comes from a named, current source. Nothing here is invented, estimated from memory, or borrowed from a wellness blog. Where two Indian authorities disagree, both positions are shown rather than one being quietly chosen.

- ICMR-INDIAB national study, published in The Lancet Diabetes & Endocrinology (2023): prevalence of diabetes 11.4%, prediabetes 15.3%, and the 2021 projections of 101 million and 136 million.
- ICMR-INDIAB-13, The Lancet Diabetes & Endocrinology (2022): achievement of ABC targets, 7.7% combined, and self-monitoring rates.
- ICMR-INDIAB-21 dietary analysis, Nature Medicine (2025): carbohydrate quality, 62.3% of energy from low-quality carbohydrate, and the modelled effects of substituting carbohydrate with protein.
- ICMR-NIN, Dietary Guidelines for Indians 2024, including the 'My Plate for the Day' and the 45% cereal ceiling.
- RSSDI Clinical Practice Recommendations for the Management of Type 2 Diabetes Mellitus 2022: diagnostic criteria, the caution on HbA1c as a sole criterion in India, ABC targets, monitoring intervals.
- ICMR Standard Treatment Workflow for Diabetes Mellitus Type 2: metabolic targets, therapy thresholds by HbA1c, the annual monitoring set, and the gestational diabetes thresholds.
- Indian Diabetes Prevention Programme, IDPP-1, Diabetologia (2006): 55% against 39.3% progression over three years, 28.5% relative risk reduction, number needed to treat 6.4.
- Sports Medicine (2022) meta-analysis on interrupting prolonged sitting, and Sports Medicine (2023) meta-analysis on pre- versus post-meal exercise.
- IRDAI Master Circular on Health Insurance Business (May 2024) and the Insurance Products Regulations 2024: the 36-month pre-existing disease waiting period and the 60-month moratorium.
- Ministry of Health and Family Welfare and PIB releases on NP-NCD, Ayushman Arogya Mandir and PM-JAY, including screening coverage to 2025 and the AAM portal count as of November 2026.
- Reuters, CNBC and The Hindu BusinessLine reporting on the March 2026 semaglutide patent expiry, generic launches and the April 2026 innovator price cuts; CDSCO Subject Expert Committee minutes as reported in October 2026 on the oral Wegovy application.

Facts were verified in November 2026. Medicine prices, scheme details and regulatory positions change; check the current position with your doctor or pharmacist before acting on any price or product detail in this book.

One last thing, and it is the most important page

This is an educational playbook, not medical advice, and it cannot replace a doctor who has examined you. Diabetes is a condition where individual targets, medicine choices and doses depend on your age, your kidneys, your heart, your weight, your other conditions and your life. Use this book to understand your numbers, to change what is on your plate, to walk after meals, to know which tests you are owed, and to ask better questions. Then take those questions to a qualified doctor, and follow their plan. That combination, an informed patient and a competent clinician, is what actually changes outcomes.